On Gaussian Interference Channels with Mixed Gaussian and Discrete Inputs

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OUTLINE

- CHANNEL MODEL AND PAST WORK
- ADVANTAGES OF MIXED INPUTS (MI)
- MUTUAL INFO. LOWER BOUND WITH MI
- CONNECTION TO SUM-SET THEORY
- MAIN RESULT: TIN IS GDOF OPTIMAL
- APPLICATION TO BLOCK ASYNCHRONOUS AND CODEBOOK OBLIVIOUS GAUSSIAN IC

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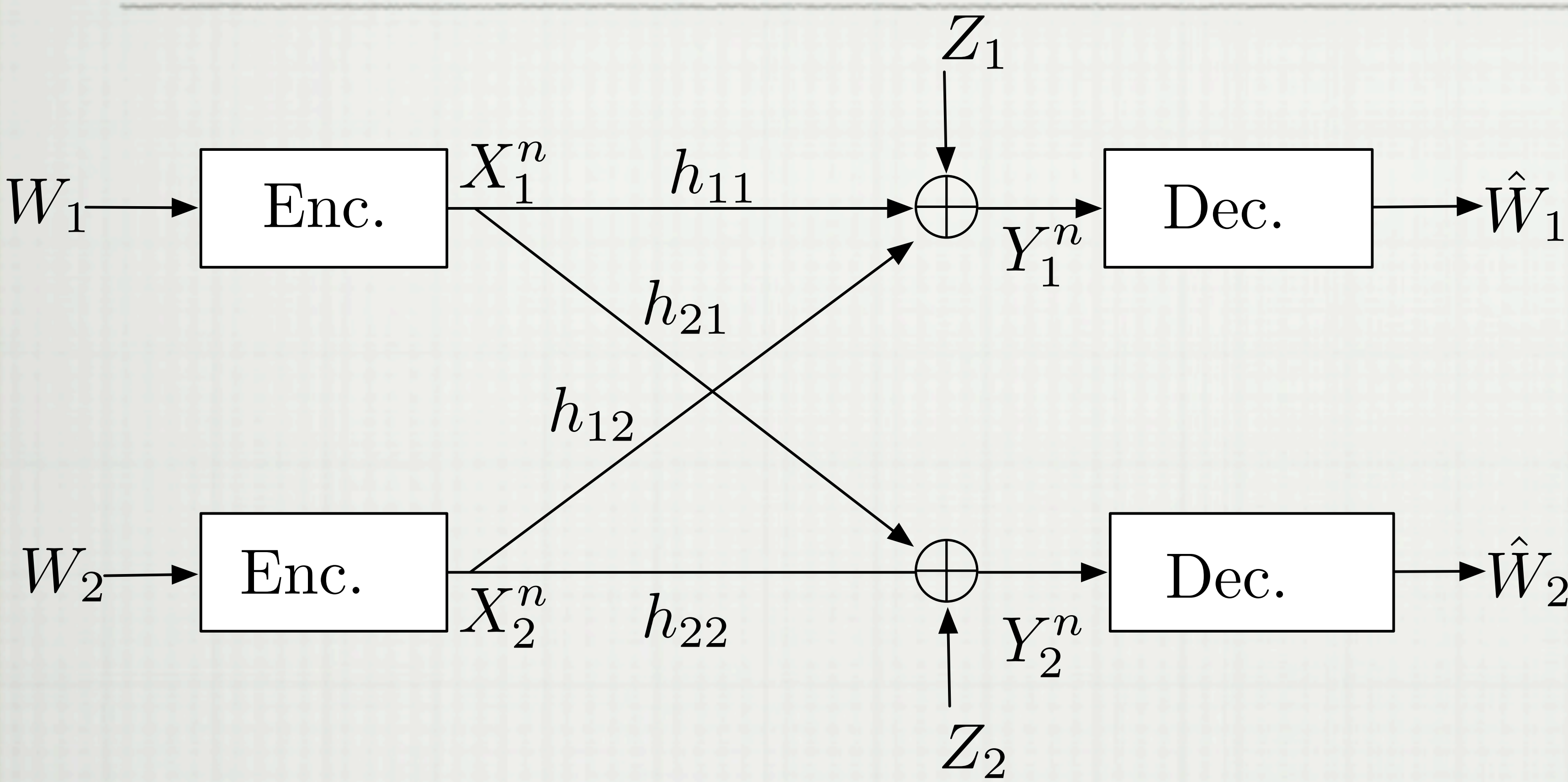
MAIN CONTRIBUTION

- SIMPLE ACHIEVABLE SCHEME: IID INPUTS AND TREAT INTERFERENCE AS NOISE (TIN)

GAUSSIAN INPUTS ARE "BAD" IN TIN REGION SOMETIME OUTPERFORMED BY OTHER INPUTS

- BY CAREFULLY "MIXING" DISCRETE AND GAUSSIAN INPUTS WE SURPRISINGLY SHOW GDOF OPTIMALITY OF TIN

CHANNEL MODEL



SYMMETRIC:

$$|h_{11}|^2 = |h_{22}|^2 = \mathrm{S}$$

$$|h_{21}|^2 = |h_{12}|^2 = \mathrm{I}$$

POWER CONSTRAINT

$$\frac{1}{n}\sum_{t=1}^{n} X_{j,t}^2 \leq 1, \;\; j = 1, 2$$

KNOWN RESULTS

CAPACITY REGION

$$\mathcal{C}^{\mathrm{IC}} = \lim_{n\to\infty} \mathrm{co}\left(\bigcup_{P_{X_1^n X_2^n} = P_{X_1^n} P_{X_2^n}} \left\{ (R_1, R_2) : \begin{array}{l} R_1 \le \frac{1}{n} I(X_1^n; Y_1^n) \\ R_2 \le \frac{1}{n} I(X_2^n; Y_2^n) \end{array} \right\} \right)$$

R. Ahlswede, “Multi-way communication channels,” in Proc. IEEE Int. Symp. Inf. Theory, Tsahkadsor, Mar. 1973, pp. 23–52.

NOT CLEAR HOW TO COMPUTE IT; MULTIVARIATE GAUSSIAN IN NOT OPTIMAL.

R. Cheng and S. Verdu, “On limiting characterizations of memoryless multiuser capacity regions,” IEEE Trans. Inf. Theory, vol. 39, no. 2, pp. 609–612, 1993.

KNOWN RESULTS

CAPACITY REGION IN STRONG INTERFERENCE

$$|h_{11}|^2 \leq |h_{21}|^2 \text{ and } |h_{22}|^2 \leq |h_{12}|^2$$

Sato, H., "The capacity of the Gaussian interference channel under strong interference," IEEE Trans. Inf. Theory, vol. 27, no. 6, pp. 786,788, Nov 1981.

CAPACITY REGION TO WITHIN 1/2 BIT

R. Etkin, D. Tse, and H. Wang, “Gaussian interference channel capacity to within one bit,” IEEE Trans. Inf. Theory, vol. 54, no. 12, pp. 5534–5562, Dec. 2008.

SUM-CAPACITY IN VERY WEAK INTERFERENCE

I.I.D. GAUSSIAN OPTIMAL FOR

$$\sqrt{\frac{\mathrm{S}}{\mathrm{I}}}(1+\mathrm{I}) \leq \frac{1}{2}$$

X. Shang, G. Kramer, and B. Chen, “A new outer bound and the noisy-interference sum rate capacity for Gaussian interference channels,” IEEE Trans. Inf. Theory, vol. 55, no. 2, pp. 689–699, 2009.

Motahari, A.S.; Khandani, A.K., "Capacity Bounds for the Gaussian Interference Channel," *Information Theory, IEEE Transactions on* , vol.55, no.2, pp.620,643, Feb. 2009

Annapureddy, V.S.; Veeravalli, V.V., "Gaussian Interference Networks: Sum Capacity in the Low-Interference Regime and New Outer Bounds on the Capacity Region," *Information Theory, IEEE Transactions on* , vol.55, no.7, pp.3032,3050, July 2009

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SUM-CAPACITY

SUM-CAPACITY

$$\mathcal{C}_{\Sigma}^{\mathrm{IC}} = \sup_{n} \max_{P_{X_1^n} P_{X_2^n}} \frac{1}{n} \sum_{u=1}^{2} I(X_u^n; Y_u^n)$$

INNER BOUND WITH I.I.D. INPUTS

$$R_L = \max_{P_{X_1,X_2} = P_{X_1} P_{X_2}} I(X_1;Y_1) + I(X_2;Y_2)$$

TREATING INTERFERENCE AS NOISE (TIN).

GAUSSIAN INPUTS ARE NOT OPTIMAL.

E. Abbe and L. Zheng, "A coordinate system for gaussian networks," IEEE Trans. Inf. Theory, vol. 58, no. 2, pp. 721–733. 2012.

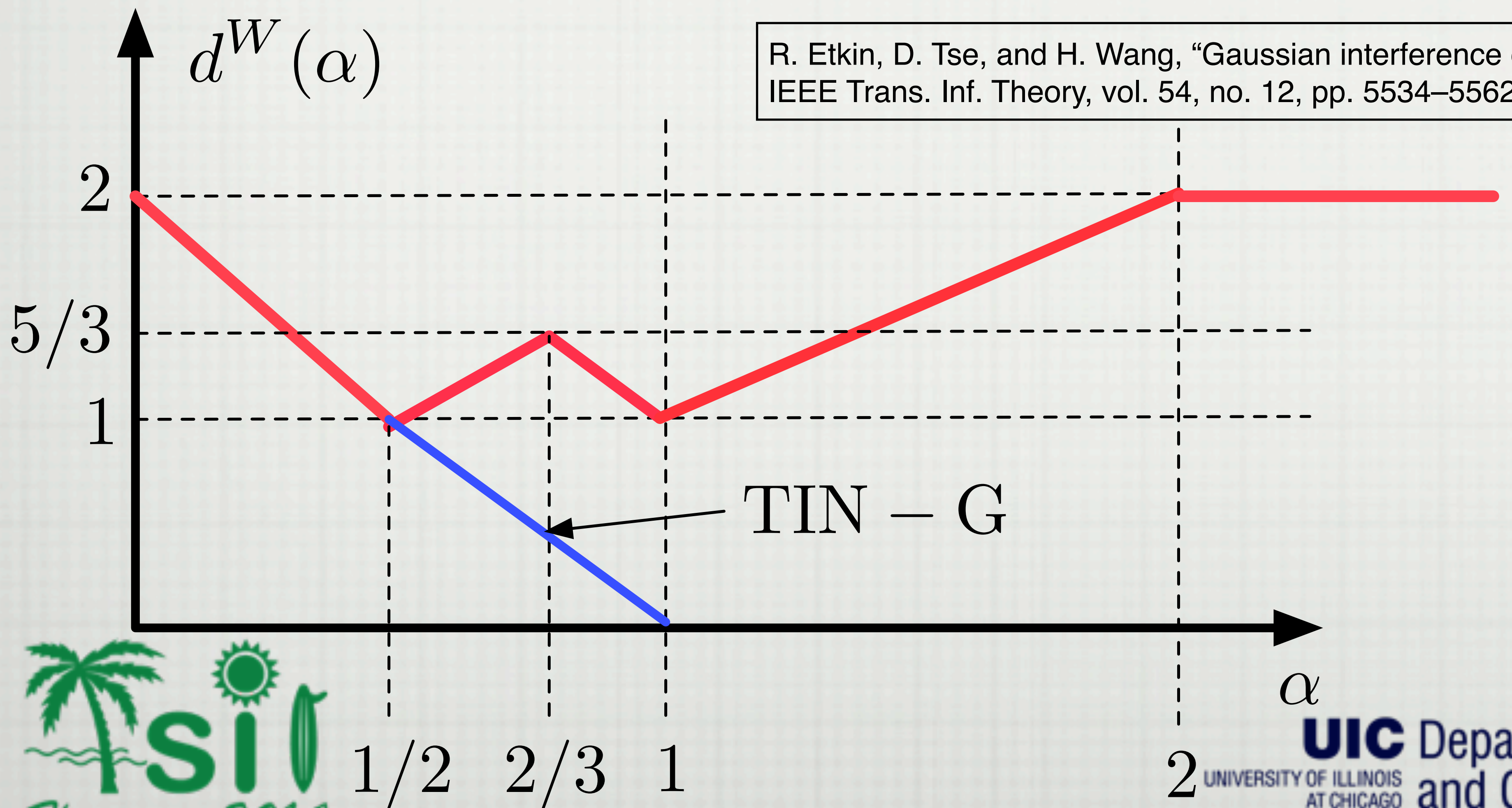
WHAT IS MAXIMIZING INPUT DISTRIBUTION?

SUM GDOF

GENERALIZED DEGREES OF FREEDOM (GDOF)

$$d(\alpha) := \lim_{\mathrm{S}\to\infty} \frac{C_\Sigma^{\mathrm{IC}}}{\frac{1}{2}\log(\mathrm{S})} : \mathrm{I} = \mathrm{S}^\alpha$$

R. Etkin, D. Tse, and H. Wang, “Gaussian interference channel capacity to within one bit,” IEEE Trans. Inf. Theory, vol. 54, no. 12, pp. 5534–5562, Dec. 2008.



MIXED INPUTS

INPUT

$$X_i = \sqrt{\delta_i} X_{G,i} + \sqrt{1-\delta_i} X_{D,i} \ : \ \delta_i \leq 1,$$

$$X_{G,i} \sim \mathcal{N}(0,1), \ X_{D,i} \sim \mathrm{PAM}(\mathrm{N}_i), \ i \in \{1,2\}$$

EVALUATE

$$R_L = I(X_1;Y_1) + I(X_2;Y_2)$$

MIXED INPUTS

DESIRABLE PROPERTIES:

$$I(X_D; \mathrm{h}X_D + Z_G) \approx \log(N)$$ **'GOOD' INPUT**

$$I(X_G; \mathrm{h}X_G + X_D + Z_G) \approx I(X_G; \mathrm{h}X_G + Z_G)$$ **'GOOD' INTERFERENCE**

Dytso, A.; Tuninetti, D.; Devroye, N., "On discrete alphabets for the two-user Gaussian interference channel with one receiver lacking knowledge of the interfering codebook," ITA, 2014 , vol., no., pp.1,8, 9-14 Feb. 2014

NEW LOWER BOUND

Theorem 1. Let X_D be a discrete r.v. with N distinct masses and minimum distance $d_{\min}$, $Z_G \sim \mathcal{N}(0,1)$. Then, for any constant h

$$I(X_D; hX_D + Z_G) \geq \left[\log(N) - \frac{1}{2}\log\left(\frac{\mathrm{e}}{2}\right) - \log\left(1 + (N-1)\mathrm{e}^{-\frac{|h|^2 d_{\min}^2}{4}}\right)\right]^+$$

where $d_{\min}^2 := \min_{s_i, s_j \in \mathrm{supp}(\mathrm{X_D}): i \neq j} |s_i - s_j|^2$.

HOLDS FOR ALL H (SNR) AND ALL INPUT DISTRIBUTIONS

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SUM SETS: CARDINALITY

$$\begin{aligned} Y_1 = h_{11}\sqrt{\delta_1}X_{G,1} + \boxed{h_{11}\sqrt{1-\delta_1}X_{D,1}} \\ + h_{12}\sqrt{\delta_2}X_{G,1} + \boxed{h_{12}\sqrt{1-\delta_2}X_{D,2}} + Z_1 \end{aligned}$$

CARDINALITY????

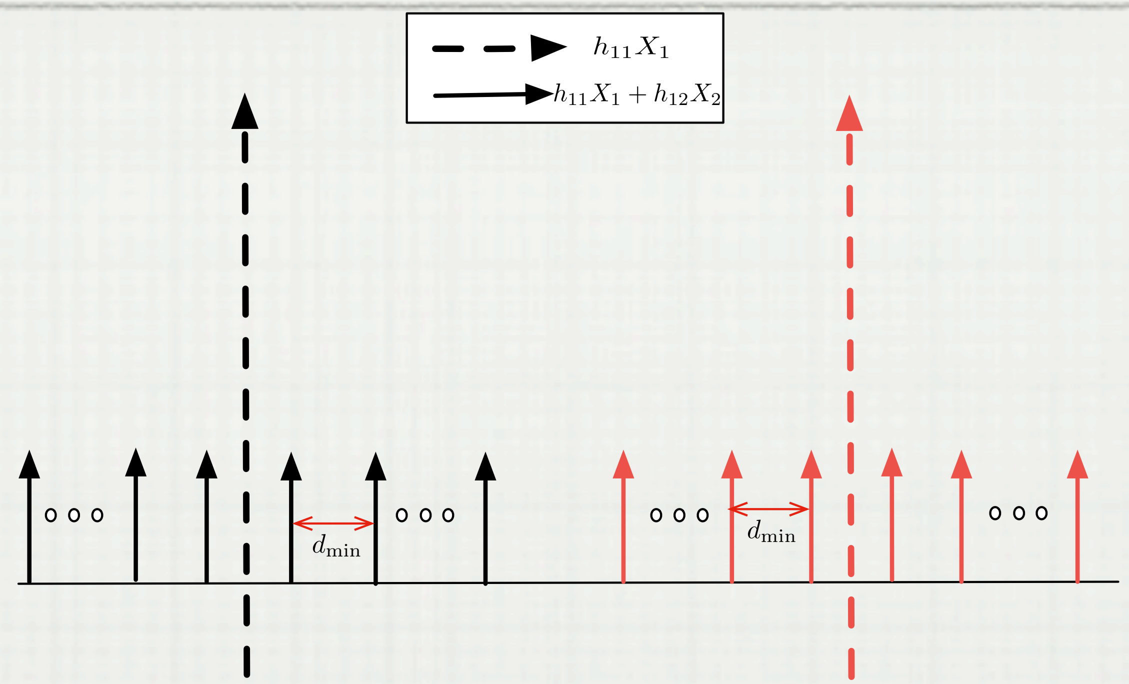
MINIMUM DISTANCE????

CARDINALITY

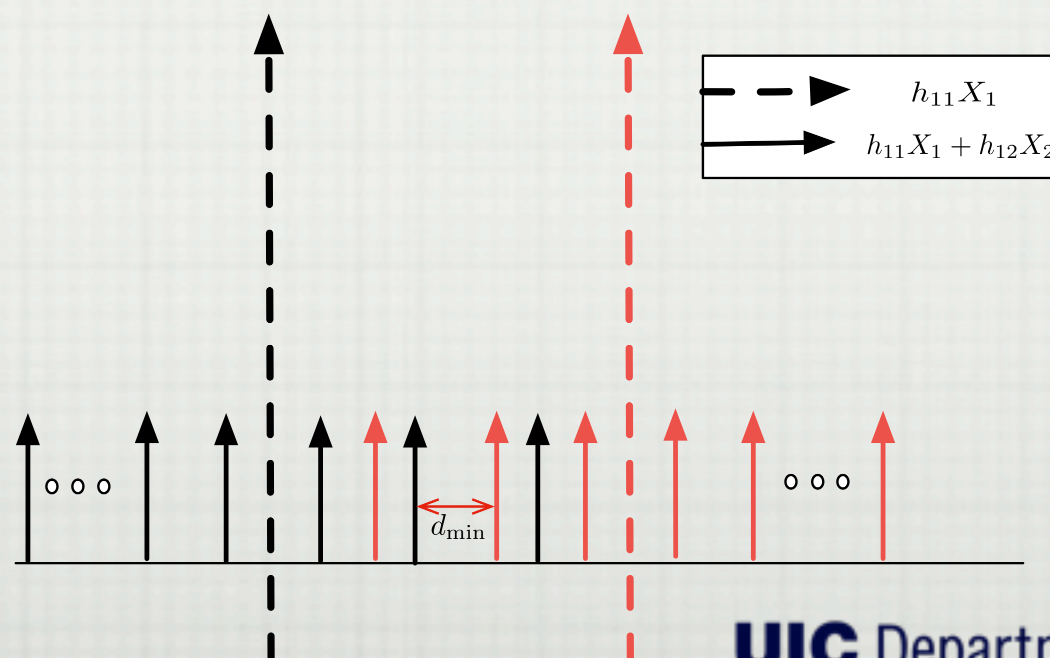
One can show show that $|\bar{h}_{11}X_{D,1} + \bar{h}_{12}X_{D,2}| = |X_{D,1}||X_{D,2}|$ almost everywhere

SUM SETS: DMIN

SCENARIO 1



SCENARIO 2



SUM SETS: DMIN

SCENARIO 1

Lemma : $d_{\min(h_{11}X_{D,1}+h_{12}X_{D,2})} = \min(|h_{11}|d_{\min(X_1)}, |h_{12}|d_{\min(X_2)})$

$$\text{if} \quad |X_{D,2}||h_{12}|d_{\min(X_{D,2})} \leq |h_{11}|d_{\min(X_{D,1})}, \tag{1}$$

$$\text{or if} \quad |X_{D,1}||h_{11}|d_{\min(X_{D,1})} \leq |h_{12}|d_{\min(X_{D,2})}. \tag{2}$$

SCENARIO 2

Lemma : $d_{\min(h_{11}X_1+h_{12}X_2)} \geq \min(1,\gamma)\min(|h_{11}|d_{\min(X_{D,1})}, |h_{12}|d_{\min(X_{D,2})})$ for all $(h_{11}, h_{12}) \in E \subset \mathbb{R}^2$ where the complement of E has measure smaller that 2γ, for any $\gamma > 0$.

SELECTING NUMBER OF POINTS

POINT-TO-POINT AWGN

$$C = \frac{1}{2}\log(1+\mathrm{SNR})$$

$$I(X_D; \mathrm{h}X_D + Z_G) \approx \log(N)$$

CHOOSE $N = \lfloor \sqrt{1+\mathrm{SNR}} \rfloor$

$$\mathrm{SNR} := |h|^2$$

FOR IC: LOG(N) = RATE COMMON MESSAGE IN HK SCHEME

RESULTS

HIGH SNR: DGOF

Theorem 1: Mixed Inputs achieve $d^W(\alpha)$ up to an outage set of measure 2γ for any $\gamma > 0$.

FINITE SNR RESULT: GAP

Theorem 2: Mixed inputs achieve sum capacity to with in $O(\log\log(\mathrm{SNR}))$ up to an outage set of measure 2γ, for any $\gamma > 0$

APPLICATIONS

BLOCK ASYNCHRONOUS IC

$$Y_{1,i} = h_{11}X_{1,i} + h_{12}X_{2,i-D_2} + Z_{1,i}$$
$$Y_{2,i} = h_{21}X_{1,i-D_1} + h_{22}X_{2,i} + Z_{2,i}$$

E. Calvo, J. Fonollosa, and J. Vidal, "On the totally asynchronous interference channel with single-user receivers," in Proc. IEEE Int. Symp. Inf. Theory, 2009, pp. 2587–2591.

IC WITH NO CODEBOOK KNOWLEDGE

$$C = \max_{P_{X_1^n, X_2^n} = P_{X_1} P_{X_2}} I(X_1; Y_1) + I(X_2; Y_2)$$

O. Simeone, E. Erkip, and S. Shamai, "On codebook information for interference relay channels with out-of-band relaying," IEEE Trans. Inf. Theory, vol. 57, no. 5, pp. 2880 –2888, May 2011.

CONCLUSIONS

- TREAT INTERFERENCE AS NOISE IS GDOF OPTIMAL IF YOU CHOOSE YOUR INPUTS SMARTLY

- CURRENT WORK: EXTENSION TO
  - WHOLE CAPACITY REGION
  - K-USER, K > 2

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MORE DETAILS

Dytso, A.; Tuninetti, D.; Devroye, N., "On the Two-user Interference Channel with Partial Codebook Knowledge at one Receiver," submitted to IT, April 2014
http://arxiv.org/abs/1405.1117

Dytso, A.; Tuninetti, D.; Devroye, N., "On the optimality of TIN in competitive scenarios," to be submitted to IT

THANK YOU

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